Task 4: Azure Managed Disks & Unmanaged Disks

Objective

To understand the difference between Azure Managed Disks and Unmanaged Disks and verify the disk type used by the Windows Virtual Machine.

Theory

Azure Managed Disk

Azure Managed Disk is a virtual hard disk service where Azure automatically manages the underlying storage accounts. It provides better scalability, availability, security, and simplified management for Virtual Machines.

Azure Unmanaged Disk

Azure Unmanaged Disk stores VM VHD files inside a user-managed Storage Account. The user is responsible for managing the storage account, capacity planning, and performance.

Comparison of Managed and Unmanaged Disks

Managed Disk	Unmanaged Disk
Azure manages storage accounts automatically	User manages storage accounts manually
Easy to deploy and manage	More administrative effort required
Better scalability and availability	Storage account limitations can affect performance
Recommended by Microsoft	Legacy storage model
Supports advanced Azure features	Limited feature support

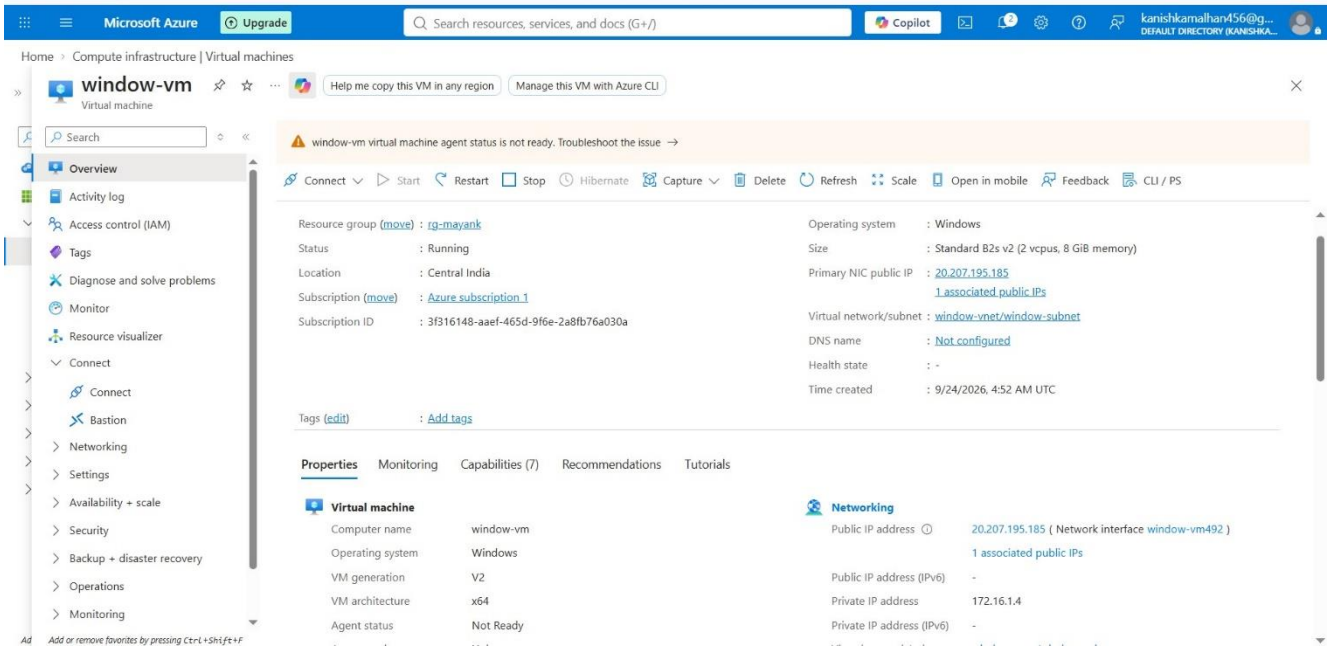
Components Used

- Microsoft Azure Portal
- Windows Virtual Machine
- Managed OS Disk

Procedure

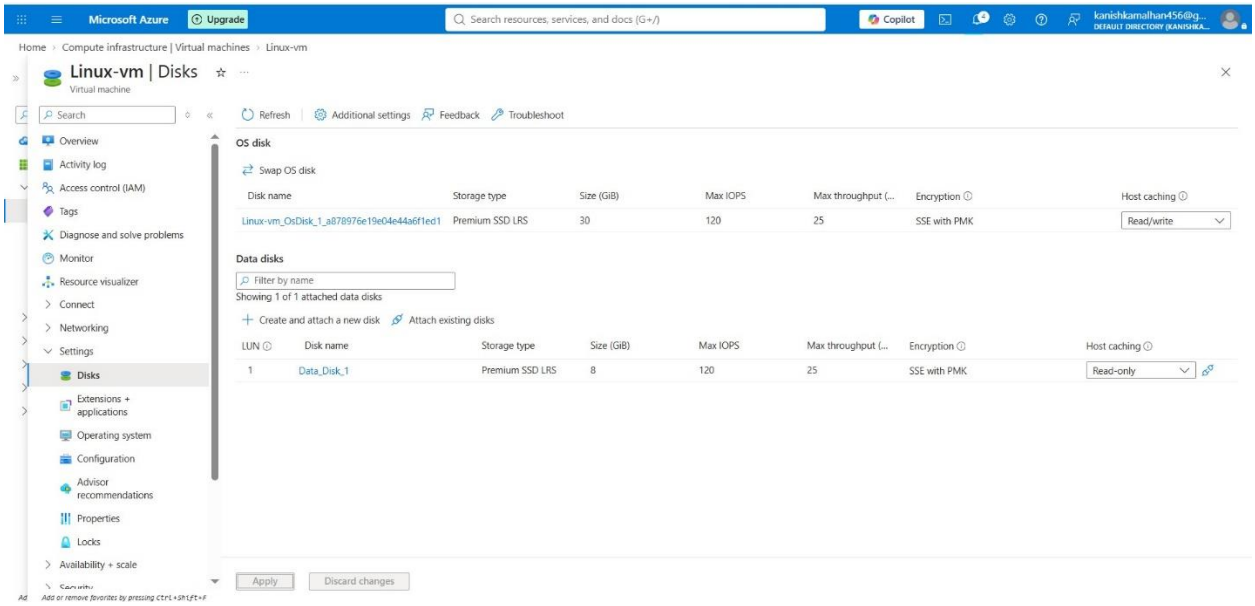
Step 1: Open the Virtual Machine

1. Sign in to the Azure Portal.
2. Navigate to **Virtual Machines**.
3. Select the Windows Virtual Machine.



Step 2: Open the OS Disk Details

1. From the VM menu, select **Disks**.
2. Click on the Operating System Disk.
3. Open the **Overview** page of the disk.

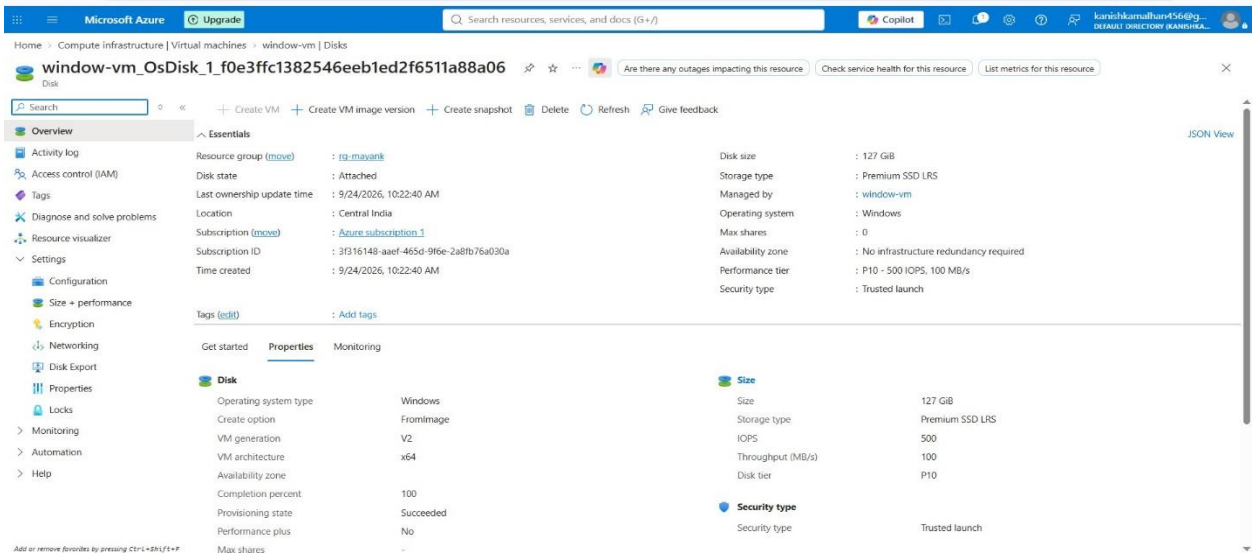


Step 3: Verify Managed Disk Configuration

In the Disk Overview page, verify the following properties:

- Disk State: Attached
- Disk Size: 127 GiB
- Storage Type: Premium SSD LRS
- Managed By: window-vm
- Operating System: Windows

The presence of the **Managed By** field confirms that the disk is configured as an Azure Managed Disk.



Observation

The Windows Virtual Machine uses an Azure Managed Disk with the following details:

- Disk Name: window-vm_OsDisk_1_f0e3ffc1382546eeb1ed2f6511a88a06
- Storage Type: Premium SSD LRS
- Disk Size: 127 GiB
- VM Name: window-vm
- Disk State: Attached

Result

The Windows Virtual Machine was successfully verified to be using an Azure Managed Disk. The disk is automatically managed by Azure, eliminating the need for manual storage account management. The differences between Managed and Unmanaged Disks were studied and understood successfully.